AI-Based Contract Risk Analysis and Clause Recommendation System

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Abstract—Contracts form the basis of almost all commercial relationships, but the manual examination of long legal documents is still a slow, expensive process and is prone to human error. Clauses that pose a risk—such as unlimited liability, automatic renewal, one-sided indemnity, or unclear terms regarding termination and payment—can lead organisations to face serious legal and financial repercussions if they are not identified. Currently available automated tools are mainly limited to keyword searching or clause-level classification, provide little scope for organisation-specific customisation, and seldom explain their decisions or suggest revised wording. This paper outlines the design of an AI-Based Contract Risk Analysis and Clause Recommendation System which takes PDFs and Word documents as input, extracts and divides the clauses, classifies them into standard legal categories, assigns a risk severity on a four-level scale (Low, Medium, High, Critical), and uses a locally hosted Large Language Model served through Ollama to generate clause recommendations that are grounded in the context of the contract and organisational guidelines. The proposed architecture integrates transformer-based clause classification (using LegalBERT/RoBERTa/DeBERTa models fine-tuned on the CUAD dataset) with a locally hosted Large Language Model served through Ollama for generating the recommendations, all made accessible via an interactive dashboard that provides risk visualisation, explanations linked to the source material, and exportable reports. The design is inspired by recent research into LLM-based legal risk assessment using Ollama and extends that work towards an end-to-end, explainable, organisation-adaptable process covering the entire pipeline from ingestion to recommendation. This paper details the present state of the prototype implementation, the modules that are still proposed or planned, and a credible evaluation methodology that will be employed to assess the system when it is fully realised, without including any invented quantitative results.

Keywords—Contract Risk Analysis, Natural Language Processing, Large Language Models, Ollama, LegalBERT, Clause Classification, Risk Severity Scoring, Clause Recommendation, CUAD, Legal AI, Local LLM, Explainable AI.

I. INTRODUCTION

Contracts form the backbone of nearly every business transaction defining the rights, obligations and risk allocation between parties. Because a contract's wording ultimately determines an organisation's exposure to financial risk careful review before signing is not optional but essential.. As the volume of commercial agreements handled by modern organisations grows, the process of reviewing them by hand struggles to keep pace.

Manual contract review is time-consuming, expensive and difficult to scale. A single agreement may run to dozens of pages of legal language and identifying the handful of clauses that carry material risk requires sustained expert attention. This dependence on reviewers introduces inconsistency: two lawyers examining the same clause may reach different conclusions about its severity and fatigue or time pressure can cause important risk indicators to be overlooked entirely.

The consequences of missing or poorly worded clauses can be severe. An unlimited liability clause can expose a company to damages beyond the value of the underlying contract. An automatic renewal clause with a cancellation window can lock an organisation into unfavourable terms for years. A one-sided indemnity clause can shift risk onto one party.. Ambiguous termination or payment conditions can create prolonged disputes. Because such clauses are often embedded in routine boilerplate language they are easy to overlook during a fast manual read-through.

In parallel the past decade has seen progress in Natural Language Processing particularly with the emergence of Transformer-based architectures BERT-family encoder models and Large Language Models. These models have demonstrated an ability to understand and generate natural language. Recently local Large Language Models served through frameworks such as Ollama have enabled organisations to run generative models privately and ground their outputs in domain-specific context rather than relying solely on static pre-trained parameters or external APIs. These advances make it increasingly feasible to automate portions of contract analysis that were previously restricted to human experts.

However existing automated approaches to contract review remain limited in respects. Rule-based and keyword-driven systems fail to generalise to paraphrased or novel clause wording. Academic systems built around BERT-style classifiers typically stop at labelling a clause's category or at its risk level without proposing an improved alternative. Commercial Contract Lifecycle Management platforms are largely closed, expensive and not customisable to a given

organisation's playbooks and risk appetite. Across both commercial systems explainability is frequently weak: end users are shown a verdict without a clear evidence-linked justification, which undermines trust and adoption.

This gap—between systems that merely classify or score clauses and the practical need for an organisation-adaptable end-to-end review assistant that also recommends improved wording—motivates the present work. The proposed AI-Based Contract Risk Analysis and Clause Recommendation System is designed to close this gap by combining clause-level classification, multi-level risk scoring and Ollama-based recommendation generation grounded in the contract context and organisational guidelines, all surfaced through an explainable dashboard-driven interface. The principal contributions of this work are: (i) an end-to-end pipeline spanning document ingestion, clause segmentation, classification and risk scoring through to recommendation generation; (ii) the use of a locally hosted Ollama LLM to generate clause recommendations guided by organisational guidelines and the surrounding contract context rather than generic model knowledge alone; and (iii) an explainability layer that links every risk verdict and recommendation to its supporting evidence presented through an interactive dashboard.

II. LITERATURE REVIEW

Research on automated contract and legal text analysis has moved forward along complementary paths: creating annotated legal datasets using transformer models for clause classification building dedicated risk-detection models and more recently using large language models (LLM), including locally hosted models served through Ollama, that can assess risk and give recommendations. A foundational contribution to this field is the Contract Understanding Atticus Dataset (CUAD) which supplies expert annotations for 510 contracts spread across 41 clause types. CUAD has become the widely used public benchmark for clause extraction and classification and it underlies much of the later work in this area including the clause-classification part of the present project. Building on datasets several studies have applied BERT-family encoders for clause-level and risk-level classification. One study fine-tuned BERT on construction specification documents to spot risk clauses reporting strong classification performance in that domain. A broader survey of contract classification gathered the field finding seven major classification task types and fourteen related datasets showing both the maturity and the fragmentation of current approaches. Complementary work used GPT-based data augmentation to tackle the class imbalance that's common in risk-labelled contract corpora reporting better handling of minority risk categories through synthetic example generation. Similarly prompt-engineered LLM approaches have shown that large pre-trained models can perform clause extraction with effectiveness even without task-specific fine-tuning highlighting the growing feasibility of few-shot and zero-shot legal natural-language-processing (NLP). A parallel line of work has moved beyond coarse classification toward graded risk severity and construction/engineering-specific review. Knowledge-graph-enhanced LLM pipelines have been applied to construction contract review improving both the accuracy and the interpretability of risk identification by linking extracted clauses to domain knowledge. In the legal context hybrid pipelines combining domain-adapted BERT variants with general-purpose and locally hosted LLMs have proven effective for analysing commercial contracts written under local legal conventions. Comparative studies of domain-trained models against frontier LLMs have further suggested that a compact specialised model can match larger general-purpose LLMs at a much lower operating cost, which is directly relevant to the design trade-offs of a deployable contract-review system. In the construction-contract domain specifically ensemble NLP pipelines that combine vectorisation with classical machine-learning classifiers have achieved strong results for tasks such as sentence-type classification and party identification though such systems remain confined to structured extraction rather than recommendation generation. Related work on text parsing has merged structure-aware segmentation with legal-domain pretrained encoders and retrieval-based evidence alignment to produce more transparent explainable risk-identification pipelines while dependency-parsing and rule-generation approaches have been used to automatically derive and continuously update risk-detection rules from contract text.

The direct technical precedent for the present project includes recent LLM-based contract review systems. Methods that combine a large language model with carefully designed prompts, false-negative reduction and dynamic context-window adjustment have demonstrated that grounding an LLM's risk verdicts and amendment suggestions in company standards and precedent significantly improves both risk-detection recall and the quality of suggested clause revisions compared with an LLM used in a purely generic fashion. Locally hosted models served through Ollama provide a practical way to obtain these benefits while keeping data and inference under organisational control.

These systems are treated as technical references for the present project as they show the practical value of generative reasoning for risk-sensitive legal text while also demonstrating the trade-off between recall-oriented risk detection and precision. Beyond pure generative pipelines multi-agent frameworks have also been proposed, in which a leader-follower (Stackelberg) game structure coordinates separate agents for risk analysis and clause revision reporting a high proportion of successfully resolved risk items while preserving legal quality. Broader technical evaluations of domain-adapted language models across classification, extraction and summarisation sub-tasks have confirmed that legal-domain adaptation yields gains over general-purpose models, on span-level and document-level legal NLP tasks.

At the core of this research are two key technical ideas. First, modern local Large Language Models (for example those served through Ollama) can be guided by carefully constructed prompts and organisational context so that their outputs remain relevant and accurate for knowledge-intensive tasks. Second the Transformer self-attention architecture is the foundation of every modern language model, such as BERT and GPT variants [17]. Another important contribution is LEGAL-BERT, a set of BERT models that were pretrained on documents. LEGAL-BERT proved that training on a domain can make a model perform better on legal natural-language-processing tasks than a general-purpose BERT, mostly for English legal text [18].

Overall this research shows three points. First transformer-based encoders work well for classifying clauses when they are fine-tuned on data sets like CUAD. Second, carefully prompted local LLMs improve both the accuracy and the relevance of risk assessment and recommendations when compared to a large-language model that is used without organisational context. Third, explainability and customisation

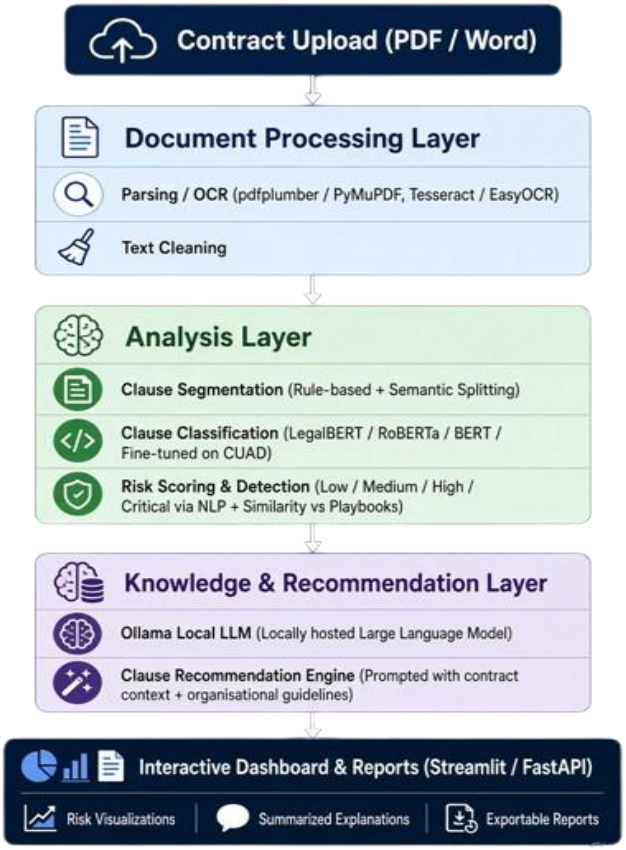
for organisations are still lacking. Most existing tools end at classifying or scoring risk. Do not provide a clear policy-based recommendation that can be acted upon.

Very few of the systems studied bring together all of the following steps, in one dashboard-driven process: splitting clauses, scoring risk severity across multiple classes generating new clause wording with a locally hosted Ollama LLM guided by organisational guidelines and contract context. The proposed AI-Based Contract Risk Analysis and Clause Recommendation System aims to fill this gap.

III. METHODOLOGY

The proposed system works through a step-by-step process. First contracts are uploaded. Then the system parses the document using OCR if needed. Next it cleans the text to remove noise and errors. After that it breaks the text into clauses. Each clause is then classified into a category. The system detects any risks in those clauses. Assigns a severity level. It prepares organisational guidelines and contract context. Using a locally hosted Ollama Large Language Model it generates recommendations. It builds explanations for those recommendations using the data it found. Finally everything is shown in a dashboard. Can be exported as a report. Each step takes the output from the one and makes it more organized and precise. This way the final recommendation can be traced back, to the clause and the exact evidence that supports it.

A. System Architecture



The design is split into three parts. The document processing part takes a contract that is uploaded and turns it into text that computers can read. The analysis part breaks down the contract into parts sorts them and checks for risks. The suggestion part uses a locally hosted Ollama Large Language Model together with organisational guidelines and contract context to come up with good suggestions for parts of the contract. These suggestions are then shown to the user through the display part. Arranging the process like this means each part can be worked on checked and made better on its own while making sure the way data moves from start, to finish stays clear and organized.

B. Document Ingestion and Pre-processing

Digital contracts are analyzed using pdfplumber and PyMuPDF. These tools take text from the documents while keeping the layout information that helps with splitting the content. Since many companies have to look at scanned documents or documents that are made of images an OCR option is used. This option is called Tesseract. EasyOCR can also be used as another choice. This OCR is used when the document does not have text that can be taken out directly. This way both documents that were created digitally and scanned PDFs well as Word files can go into the same next step of processing. Making sure this step works well is important. Any problems that happen here like letters missing style or missing parts will move along, to all the next steps of looking at the documents.

C. Clause Segmentation

Once clean text is available the contract is split into clauses. This is done using a combination of rule-based segmentation. The rules look for numbering patterns, headings and standard clause markers. When these are not present or are inconsistent semantic splitting is used. Semantic splitting groups sentences based on how related they're in topic. Clause-level processing is important. It is not about looking at the document or arbitrary paragraphs. Risk, classification and recommendation are all things that belong to clauses. A single risky sentence, in a section that is low risk must not be hidden by the surrounding text. A recommendation must be linked to a clear part of the contract language.

D. Clause Classification

Segmented clauses are sorted into groups—like Liability, Termination, Payment, Confidentiality and Governing Law—by using transformer-based encoders such as LegalBERT, RoBERTa or DeBERTa that have been fine-tuned on the CUAD dataset. The CUAD dataset supplies clause-type annotations that act as a foundation for training a classifier that can generalise across common commercial contract types such as NDAs, Master Service Agreements and standard service agreements, which make up the main focus of this project. Putting a clause into a recognised category is a step, for useful risk scoring because the risk factors and organisational policies that apply to a liability clause differ from the risk factors and organisational policies that apply to a termination clause or a payment clause.

E. Risk Detection and Severity Scoring

Each classified clause is given one of four risk severity levels—Medium, High or Critical—by looking at what the clause says and comparing it with the organisation's playbooks. This comparison uses similarity and Natural Language Inference techniques to decide how far the actual wording of the clause is from the organisation's language, for that type of clause. A classified clause that matches an approved template well is marked as risk. A classified clause that puts one side in trouble cuts out protections or changes the normal rules a lot is marked with a higher risk level. By grading clauses into these four levels reviewers can focus first on the classified clauses that are most severe instead of treating every flagged clause the same.

F. Ollama-Based Recommendation Context

The part where recommendations are created uses a locally hosted Large Language Model served through Ollama. When a clause that could be risky is found, the original clause is combined with relevant organisational guidelines and surrounding contract context. This package is given to the Ollama model together with a carefully engineered prompt. This way any advice the model gives is guided by the company's rules and the specific contract situation rather than only by general information the model learned when it was first trained. Making recommendations this way should make the advice better suited for the company that is using the system and keeps all inference under local control.

G. Clause Recommendation Generation

When a risky clause is given, along with its category and its severity and the prepared organisational guidelines and contract context, the Ollama LLM creates a revised version of that risky clause. The generation process is clearly framed as decision support. The system offers wording that aims to cut down the identified risk while staying in line with the organisational guidelines supplied in the prompt.. The result is shown to a human reviewer for judgment and approval and the system does not act as independent legal advice. This framing matters both in practice and, in ethics because contract language often has meanings that rely on a negotiation context a context that an automated system cannot fully observe.

H. Explainability and Dashboard

For every flagged clause the system shows: the clause text, the category it was assigned to the risk severity level and a clear explanation in plain language about why that severity was chosen. It also includes the policy or historical agreement excerpts that were used to make the assessment.. It provides recommended wording to replace the clause. Showing this evidence trail with each decision directly tackles the problem of explainability that many current commercial and academic systems struggle with (Section II). This approach helps reviewers understand the system's decisions and gives them the ability to audit the results. The user interface is built with Streamlit, powered by FastAPI in the background. It offers risk displays, clickable links, to the sources used in the analysis and reports that can be downloaded for offline use or shared with stakeholders who don't use the dashboard.

TABLE I

COMPARISON OF EXISTING APPROACHES AND THE PROPOSED SYSTEM

Dimension	Manual	Rule/KW	Proposed
Scalability	Low	Medium	High
Handles paraphrased clauses	Yes	No	Yes (class.+Ollama)
Org.-specific customisation	Yes (manual)	Limited	Yes (Ollama)
Explainable verdicts	Yes (implicit)	Partial	Yes (evidence)
Generates improved wording	Yes (manual)	No	Yes (Ollama LLM)

TABLE II

PROJECT MODULES, TECHNOLOGIES, AND PURPOSE

Module	Technology / Purpose
Ingestion	pdfplumber, PyMuPDF, Tesseract/EasyOCR — convert input to clean text
Segmentation	Rule-based + semantic splitting — divide into clauses
Classification	LegalBERT/RoBERTa/DeBERTa (CUAD fine-tuned) — assign clause categories
Risk Scoring	NLI/similarity vs. playbooks — assign Low/Med/High/Critical
Retrieval	Ollama — retrieve policies and historical agreements
Recommendation	Ollama local LLM — generate grounded wording
Dashboard	Streamlit/FastAPI — visualise, explain, export reports

I. Evaluation Strategy

Because the classification, risk-scoring and recommendation components of the system are at stages of implementation the evaluation strategy is built to be applied step, by step as each component matures. The evaluation strategy does not rely on an end-to-end measurement that is available today. Section IV.B explains the evaluation dimensions that will be used for each pipeline stage once the corresponding models are trained and linked to live data.

IV. RESULTS AND ANALYSIS

A. Current Implementation Status

The current stage of the project is a working end-to-end prototype that shows the pipeline in theory with each module at a different level of development. Document ingestion, clause segmentation, risk-score presentation and the dashboard/reporting interface have been shown in a form using sample contracts proving that the overall pipeline design and the flow of information between stages are logically sound. The CUAD-based clause classification model, the Ollama-based recommendation engine are, at this stage proposed components. Their design, expected inputs and outputs and integration points within the pipeline have been written down. They have not yet been trained, connected to live data or tested end-to-end. This separation is kept deliberately throughout this paper.

Qualitative claims about what the system's intended to achieve – automated clause analysis, risk-aware review policy-grounded recommendations and reduced manual review effort – describe the project's design goals and the behaviour shown by the mocked prototype not results from a fully trained and deployed system. Within this scope the project has produced a demonstrable workflow. A contract can be uploaded, split into clauses shown against a four-level risk scale with placeholder scoring logic and displayed with text and a mock recommendation all through the intended dashboard interface. This confirms the feasibility of the proposed architecture and gives a solid base, for linking the remaining trained-model components.

TABLE III

PROPOSED EVALUATION METRICS BY PIPELINE STAGE

Stage	Proposed Metrics
Classification	Accuracy, Precision, Recall, F1, Macro-F1
Risk Detection	Precision, Recall, F1, false-negative rate
Risk Severity	Expert agreement, weighted F1, confusion matrix
Ollama Recommendation	Recommendation relevance, usefulness, expert agreement
Recommendation	Expert agreement, semantic similarity, usefulness
Efficiency	Processing time, review-time reduction, scalability

B. Proposed Evaluation Framework

Because the classification, risk-scoring and recommendation components have not yet been trained and evaluated against ground truth this paper does not report any experimental results. Instead the authors propose evaluation dimensions that will be used once the classification, risk-scoring and recommendation components are finished. These dimensions are based on evaluation practices that were discussed in Section II and include recall-oriented and amendment-quality metrics that were used for LLM-and-Ollama-based risk assessment in the base reference system [13]. Conceptually the proposed system can be placed on a spectrum of existing contract-review approaches a spectrum summarised in Table I. The proposed system wants to keep the interpretability that comes from manual review while getting closer to the scalability that automated tools offer. It also wants to do than rule-based and classification-only NLP systems by adding organisation-specific grounding and generative recommendation. The authors present this placement as a design rationale than a claim of measured superiority because no comparative experiments, against these other approaches have been carried out yet.

C. Discussion

The design has a few benefits when it is fully put into action. Making suggestions with a locally hosted Ollama model guided by a company's rules and the specific contract context instead of just using what an AI model knows in general is expected to make the ideas more suitable for the situation and more likely to be agreed upon by legal people. The clear four-level risk scale, along with explanations linked to the source is meant to help with sorting tasks quickly by focusing on the most important issues first. The design is also made up of parts that can be changed on their own as new types of clauses, rules or company policies come up without having to rebuild the whole system from scratch.

At the time there are some things that are not good about the design. The quality of the risk evaluation and the suggestions depends on the quality. How complete the company's policy documents and past agreements are. If these are not good or are out of date the system will not be very helpful no matter how good the model is. Datasets like CUAD are useful. They are mostly made with English commercial contracts and might not work perfectly for all kinds of clauses or places the system might need to handle. Like any AI-based part there is a chance of made-up or not supported words in the suggestions. That is why the part that explains things clearly and having a person check the work are required, not just optional. Local LLM inference with Ollama requires adequate local compute resources and this needs to be balanced with how fast the system should work in the real world. Lastly because some of the parts are still in the design phase or just mockups the claims in this paper, about how well the system works should be seen as goals that are supported by the way the system is built and what other research has shown. These claims are not proven yet. Need the tests described in Section IV.B to be done.

V. CONCLUSION AND FUTURE WORK

Manual review of contracts takes a long time costs a lot of money and often leads to inconsistent results. Automated tools available today only go as far as classifying clauses or work inside closed systems that cannot be customized. These tools also lack explanations for their outputs. This paper introduces an AI-based system designed to handle contract risk analysis and suggest improvements in clauses. The goal is to fill the gap between manual reviews and limited automation.

The system uses key technologies. It starts by breaking down contracts into clauses then uses machine learning models based on transformers to sort these clauses into standard legal categories. Next it assigns a risk level using a four-tiered severity scale. To make suggestions meaningful it uses a locally hosted Large Language Model served through Ollama, guided by organisational guidelines and the surrounding contract context. The full process includes document upload, clause segmentation, classification, risk scoring, retrieval, recommendation generation and a dashboard that shows everything clearly with explanations. The end result is a tool that supports teams instead of replacing them.

Now the system is still in prototype form. It runs the pipeline but in a simulated way. Components like the CUAD-based classification model and the local Ollama large language model that generates recommendations have not been fully trained or connected yet. These parts will be. Evaluated using metrics described in Section IV.B. The system is not ready for production use.. It does provide a solid foundation built on current research and sound design principles.

Future work will focus on upgrades. First adding support for languages would make the system useful for companies working across borders. Real-time collaboration features, where multiple reviewers can discuss and approve changes together would better match how teams actually work. Improving explainability is another priority—giving confidence scores for each risk assessment and recommendation would help users trust the system. Connecting the tool to existing platforms like enterprise document managers or Contract Lifecycle Management systems would allow it to fit into workflows without disrupting them. A feedback loop that learns from user actions—such as which recommendationsre accepted or ignored—would let the system improve over time. In the run a mobile app could let people review contracts anywhere they go.. Fine-tuning the models for specific industries, like IT healthcare or construction would ensure the system understands sector-specific risks and regulations..

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